CareLoop

Post-Discharge Recovery & Patient Safety Guardian

DSH HACKS V2 · AI X HEALTHCARE

Tournament Submission · Oct 2026

1 The Critical Clinical Problem

The 30 days following hospital discharge represent the highest-risk vulnerability window in the continuum of care:

- **Discharge Summary Disconnect:** Jargon-dense medical notes leave patients and families confused regarding crucial regimen adjustments and warning signs.
- **Medication Hazards:** Complex changes trigger duplicate therapy, missed critical doses, and lethal OTC drug-drug clashes.
- **Silent Deterioration:** Congestive volume creep or surgical site infection develops days before an avoidable, high-cost readmission.

2 The Solution: Four Continuous Care Loops

LOOP 1 · DECODE — **Discharge NLP Parser & Live Sandbox**

Extracts unstructured summaries across 6 languages with dual reading modes (Simple ↔ Standard ICD-10), Live Note Sandbox re-parsing, 5 quick clinical presets, and verbatim source-span traceability.

LOOP 2 · PROTECT — **Medication Guardian**

Reconciles active prescriptions against OTC products. Deterministically catches fatal interactions (e.g. NSAID + Anticoagulant, Triple Whammy AKI) before ingestion via a 30-pair clinical matrix.

LOOP 3 · WATCH — **60s Check-in & Rules**

Monitors daily dry weights, vitals, and surgical signs. Computes an explainable 0–100 Recovery Risk Index with transparent factor attribution (Green / Amber / Red). Time Machine tracks trajectory Day 0–3.

LOOP 4 · CONNECT — **Caregiver Pass & SBAR**

Generates cryptographically signed offline QR passes (/pass/token) for secure family surveillance and produces instant standardized SBAR medical handovers for outpatient clinical reviews.

3 Verified Clinical Evidence

- **1 in 5 Medicare Patients Readmitted (19.6%):** Jencks SF et al. *Rehospitalizations among patients in Medicare fee-for-service.* N Engl J Med. 2009;360:1418-1428.
- **50% Medication Discrepancies Post-Discharge:** Kripalani S et al. *Deficits in communication and information transfer.* JAMA. 2007;297(8):831-841.
- **Preventable Heart Failure Readmissions:** Ponikowski P et al. *2016 ESC Guidelines for acute and chronic heart failure.* Eur Heart J. 2016;37:2129-2200.

Safety-First Deterministic Architecture

Deterministic Symbolic NLP: <5ms entity extraction, medication dose locking, and multilingual decoding with 100% verbatim source provenance.

Deterministic CDS Engine: 100% of risk scoring, 30-rule drug interaction checks, and emergency alerts run in auditable code (FDA SaMD Class II aligned).

RECOVERY TIME MACHINE · EARLY DETERIORATION CATCH — RED ALERT (100/100)

- **Day 0–1:** Green (68.0 kg) · Baseline stable post-discharge recovery established
- **Day 2:** Amber (68.9 kg) · Missed morning diuretic dose flagged by Medication Guardian
- **Day 3:** **RED (70.4 kg, +2.4 kg in 3 days)** · Orthopnea & pedal edema detected. System auto-dispatches family SMS alert and generates Clinician SBAR prior to acute ED admission.

4 Key Technical Differentiators

- **Live Note Sandbox & Presets:** Test clinical NLP live on arbitrary pasted notes or 5 quick presets (Post-MI DAPT, COPD, HFrEF, Ortho, Diabetes).
- **Zero-Dependency & Zero Cloud Egress:** Operates 100% offline client-side with zero external API keys, zero network requests, and zero PHI leakage.
- **100% Provenance Traceability:** Every care plan recommendation links to verbatim character spans in the original hospital discharge summary.
- **Live Verification Engine:** Live streaming benchmark harness: 20 clinical discharge cases (100.0% F1) + 30 deterministic safety rule scenarios.

5 Pilot Roadmap & Clinical Impact

Phase 1 (Complete): Zero-dep app with 68 drugs, 30 interactions, 6 languages, live sandbox, clinical presets, QR caregiver pass, Time Machine, SBAR.

Phase 2 (Hospital EHR Pilot): SMART-on-FHIR connector to ingest discharge summaries directly from Epic and Cerner EHR platforms.

Projected Clinical Impact: Target 25–35% reduction in preventable 30-day readmissions through proactive early outpatient intervention.

100.0%	30 / 30	68 Drugs · 30 Pairs	0 bytes
DISCHARGE F1 BENCHMARK (20/20)	SAFETY RULES SUITE PASSED	INTERACTION MATRIX COVERAGE	CLIENT PHI EGRESS (OFFLINE)